

COLUMBUS, GA, USA

WMO: 722255

Lat: 32.516N

Lon: 84.942W

Elev: 120

StdP: 99.90

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-1.4	-14.2	1.1	1.1	-11.8	1.4	3.3	9.3	9.5	8.8	9.5	2.4	330	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	35.6	23.6	34.5	23.5	33.5	23.4	25.7	31.8	25.2	31.1	24.7	30.3	3.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.3	27.9	23.6	18.7	27.4	23.2	18.2	26.9	79.2	31.8	77.1	31.0	75.3	30.5	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.2	DB	-7.1	37.1	2.4	1.7	-8.9	38.3	-10.3	39.2	-11.7	40.2	-13.4	41.4	(4)
(5)			WB	-8.7	26.5	2.3	0.6	-10.3	27.0	-11.7	27.3	-13.0	27.7	-14.6	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	8.8	11.1	14.7	18.7	23.2	26.5	28.0	27.7	25.0	19.5	13.9	10.3		
	DBStd	7.79	5.29	5.01	4.58	3.57	3.05	2.10	1.74	1.86	2.77	3.96	4.37	4.86		
	HDD10.0	216	88	43	13	1	0	0	0	0	1	13	58			
	HDD18.3	1106	298	206	127	38	4	0	0	0	1	34	144	253		
	CDD10.0	3497	50	73	159	262	408	496	557	548	452	296	130	66		
	CDD18.3	1345	1	4	16	49	153	246	298	290	202	72	11	2		
	CDH23.3	12984	3	17	139	457	1443	2416	3147	2982	1792	523	61	6		
(13)	CDH26.7	5504	0	1	18	99	543	1101	1492	1379	736	128	6	0		
(14)	Wind	WSAvg	2.6	2.8	2.9	3.0	2.9	2.6	2.4	2.3	2.2	2.5	2.4	2.4	2.6	(14)
(15) Precipitation	PrecAvg	1258	114	115	147	103	95	102	132	98	80	55	93	115		
	PrecMax	1860	212	239	338	289	200	275	336	257	160	206	295	238		
	PrecMin	670	18	30	14	2	6	13	52	20	6	0	20	40		
	PrecStd	257	51	55	72	66	52	62	64	57	41	44	56	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	25.5	28.7	30.7	34.3	36.3	37.0	37.2	35.6	31.8	27.5	24.1		
		MCWB	17.9	18.2	18.0	20.1	22.1	23.5	23.9	23.9	22.8	22.6	19.1	18.1		
	2%	DB	21.2	23.1	26.6	29.2	32.5	34.7	35.5	35.4	33.9	29.6	25.0	22.0		
		MCWB	16.4	16.5	17.3	19.2	21.7	23.2	23.8	23.9	22.5	20.5	18.2	17.4		
	5%	DB	19.2	21.3	24.8	27.8	31.2	33.5	34.3	34.0	32.3	28.1	23.2	20.1		
		MCWB	15.3	15.6	16.3	18.5	21.3	23.0	24.0	23.8	22.2	19.7	17.2	16.7		
	10%	DB	17.3	19.3	22.8	26.2	29.8	32.1	33.1	32.7	30.9	26.4	21.5	18.2		
		MCWB	14.2	14.4	15.4	17.6	20.6	22.8	23.9	23.7	22.1	19.0	16.5	15.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	19.9	20.5	22.3	24.4	25.9	26.6	26.3	25.0	24.1	22.1	20.2		
		MCDB	21.5	23.2	24.4	27.3	30.7	32.5	32.8	32.6	30.5	28.4	24.0	22.0		
	2%	WB	17.9	18.5	19.2	21.2	23.5	25.2	25.7	25.6	24.4	22.9	20.4	18.8		
		MCDB	20.2	21.7	23.2	26.0	29.1	31.3	31.8	31.5	29.6	26.8	22.7	20.5		
	5%	WB	16.5	17.3	18.2	20.3	22.7	24.5	25.2	25.1	23.9	22.0	19.0	17.3		
		MCDB	18.6	20.0	22.4	24.8	28.1	30.3	31.2	30.8	28.8	25.3	21.9	19.5		
	10%	WB	14.8	15.8	17.0	19.4	21.9	24.0	24.7	24.7	23.3	21.0	17.6	15.8		
		MCDB	17.1	18.4	21.3	24.0	27.4	29.4	30.3	30.0	27.9	24.5	20.6	18.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.3	12.0	12.2	11.4	10.2	10.0	9.7	10.0	11.1	11.6	10.3		
		MCDBR	12.1	12.4	14.0	13.7	12.6	12.0	11.4	11.4	12.1	12.4	12.6	11.1		
		MCWBR	7.9	7.3	6.3	5.5	4.1	3.4	3.0	3.1	3.6	5.0	6.9	7.3		
		MCDBR	9.9	10.1	11.0	10.5	10.5	10.0	9.9	9.7	9.2	8.8	9.6	9.0		
	5% WB	MCWBR	7.8	7.5	6.4	5.4	4.1	3.6	3.3	3.2	3.4	4.8	6.6	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.326	0.354	0.387	0.439	0.481	0.519	0.522	0.425	0.374	0.349	0.327		
	taud		2.520	2.508	2.449	2.345	2.242	2.179	2.083	2.067	2.327	2.447	2.461	2.523		
	Ebn at Noon		898	932	925	903	854	814	781	773	842	866	858	874		
	Edh at Noon		85	95	108	125	140	149	163	164	121	100	90	81		
(44) All-Sky Solar Radiation	RadAvg		2.83	3.41	4.51	5.64	6.24	6.12	6.03	5.47	4.82	4.13	3.17	2.43		
	RadStd		0.16	0.40	0.38	0.40	0.48	0.48	0.37	0.33	0.44	0.52	0.28	0.26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	N/A		
(47)	Regional (42 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+107		

Nomenclature: See separate page